variables remain unchanged. If we condition on B, however, the variable that is correlated to it, A, gets updated.

2.2.3 Modeling many variables with high-dimensional Gaussian distributions

A multivariate Gaussian distribution need not only contain three random variables; it can in fact simultaneously model any finite number of variables. In this subsection, we will see that a higher-dimensional Gaussian works in the same way as what we have seen so far. So, let's say instead of a three-dimensional Gaussian representing three houses, we have a twenty-dimensional Gaussian encoding information about an array of houses on a street. An even higher-dimensional Gaussian would model houses in a city or country.

Further, with these parallel coordinates plots, we can visualize all individual variables of a high-dimensional Gaussian at the same time. This is because each variable corresponds to a single error bar, occupying only one slot on the *x*-axis.

We once again normalize its mean vector to be the zero vector, and while it's not convenient to show its twenty-by-twenty covariance matrix, we could plot a heat map visualizing this matrix as in Figure 2.10.

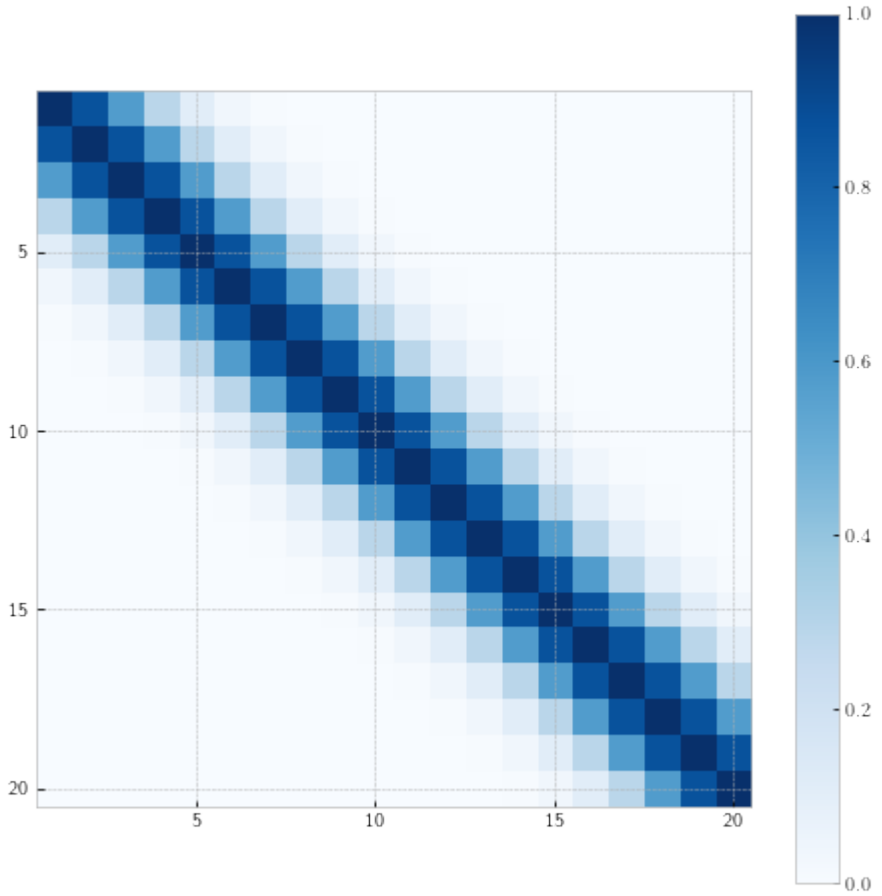


Figure 2.10 Heat map showing the covariance matrix of a twenty-dimensional Gaussian distribution. Neighboring variables are more correlated than those that are far away, as indicated by a darker shade.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>